

Yun Cheng Lin

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Education

Massachusetts Institute of Technology (MIT)

Bachelor of Science in Artificial Intelligence and Decision-Making

Cambridge, MA

June 2026

GPA: 4.5/5

Relevant Coursework: Design and Analysis of Algorithms, Software Construction, Machine Learning, Design for the Web

Skills

Languages: Python, JavaScript/TypeScript, HTML, CSS

Tech Stack: Git, React, Vue, Express, Socket.io, Supabase, Tailwind CSS, Phaser, Vercel, SQL

Relevant Experience

MIT Computer Science Undergraduate Lab Assistant

6.100A & 6.100B

Cambridge, MA

Sep 2024 – May 2026

- Assisted students in developing computational thinking skills and debugging techniques in introductory programming courses
- Guided students through concepts including dynamic programming, graph algorithms, and data science applications
- Supported problem-solving and code optimization in real-time lab environments

Projects

Multiplayer Blackjack Web Game

TypeScript, React, Express, Socket.io, Supabase, Tailwind CSS

- Designed and built a full-stack multiplayer Blackjack game with real-time game rooms (up to 3 players), including the core game logic for betting, splitting, doubling down, and dealer AI
- Designed a versioned full/patch state-sync protocol over Socket.io so multiple clients stay consistent in real time and can reconcile after disconnects
- Integrated Supabase (Postgres) for persistent user accounts, balances, and game history, with an in-memory caching layer to reduce database calls
- Designed and built a responsive React UI, including a stats/history dashboard, card-dealing animations, and dark/light theming

Farm Sprint

JavaScript, Phaser, React

- Designed and developed the quest-based game logic, where players must complete tasks before their quest board overflows
- Built game mechanics and rendering using the Phaser HTML5 game engine in JavaScript
- Integrated third-party game assets from itch.io for visuals and animations
- Designed game scenes, terrain, and UI (HUD and menus) using itch.io assets and custom assets built in Tiled

Linear Gradient Demo

JavaScript, Vue, HTML, CSS

- Developed an interactive tutorial website to teach and visualize the CSS linear-gradient function
- Designed and built the guided tutorial logic, including progressive levels, input validation, and feedback
- Implemented a playground feature enabling live preview and manipulation of gradient parameters
- Designed and engineered a dynamic UI using Vue.js components and reactive state management for user interaction

Neural Radiance Field (NeRF)

Python, Machine Learning, Computer Vision

- Implemented a Neural Radiance Field (NeRF) model in Python from the ground up, based on the original NeRF paper, to synthesize novel views of a 3D object from a sparse set of labeled 2D images
- Built a multi-layer perceptron to learn a continuous volumetric scene representation and applied volume rendering techniques to reconstruct unseen viewpoints
- Trained the model for 2,000 iterations on a drums dataset sourced from Kaggle, demonstrating accurate novel-view synthesis
- Developed and documented the project in a Google Colab notebook for reproducibility

Chat App

Vue, HTML, CSS

- Designed and built a messaging web app for MIT's 6.4500: Design for the Web, addressing the challenge of following multiple threads in large group chats
- Implemented branching subchats that let users fork a conversation into a separate thread without losing the context of the parent conversation
- Built the frontend with Vue.js components and the Graffiti API as a backend for storing user profiles and chat/message histories, using reactive state management across component, page, and store modules